Tutorial 2

MAT3007 Optimization, 2023 Summer

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6.5-6.9

Outline

1 Linear Programming and Its Standard Form

2 Reformulate Problems as LP

Definition: Linear Programming

Linear programming (LP) is an optimization problem in which the objective function and all the constraint functions are **linear**.

$$
\begin{aligned}
\text{minimize} \quad & c^\top x \\
\text{subject to} \quad & A_1 x \geq b \\
& A_2 x \leq d \\
& A_3 x = e \\
& x_i \geq 0 \quad \forall i \in N_1 \\
& x_i \leq 0 \quad \forall i \in N_2 \\
& x_i = 0 \quad \forall i \in N_3
\end{aligned}
$$

- The variable x is a n dimensional column vector.
- b, d, e are column vectors of dimension m_1,m_2,m_3, respectively.
- A_1,A_2,A_3 are matrices with dimensions $m_1 \times n$,$m_2 \times n$,$m_3 \times n$ respectively.

Standard form of LP

The standard form of LP is of the form

$$\begin{aligned} \text{minimize} \quad & c^\top x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

- The standard form formulation is helpful when implementing particular LP algorithms.
- Any LP can be transformed into standard form.

Transform LP to Standard Form

- Objective is **maximization**: transform into minimization by using $-c$.
- Constraints are **inequality**: transform into equality by adding slack variables:

$$Ax \leq b \Longrightarrow Ax + s = b, s \geq 0$$
$$Ax \geq b \Longrightarrow Ax - s = b, s \geq 0$$

- Decision variables are **non-positive** or **"free"**: introducing new variables:

$$x_i \leq 0 \Longrightarrow y_i \geq 0, y_i = -x_i$$
$$x_i \text{ "free" } \Longrightarrow x_i = x_i^+ - x_i^-, x_i^+ \geq 0, x_i^- \geq 0.$$

Exercise 1

Consider a general linear programming problem

$$
\begin{aligned}
\max_{x} \quad & \sum_{j=1}^{n} c_j x_j \\
s.t. \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i \in I \\
& \sum_{j=1}^{n} a_{ij} x_j = b_i, \quad i \in E \\
& x_j \geq 0, \quad j \in S
\end{aligned}
\tag{1}
$$

Show that it is possible to equivalently formulate it into the standard form.

Solution to Exercise 1

- For $j \notin S$, x is free, thus we must introduce new variables:

$$x_j = x_j^+ - x_j^-, x_j^+ \geq 0, x_j^- \geq 0 \tag{2}$$

- For the objective function:

$$\max_x \sum_{j=1}^{n} c_j x_j \iff \min_x \sum_{j \in S} -c_j x_j + \sum_{j \notin S} -c_j (x_j^+ - x_j^-) \tag{3}$$

- For the constraints, we do the following:

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \iff \sum_{j \in S} a_{ij} x_j + \sum_{j \notin S} a_{ij} (x_j^+ - x_j^-) + s_i = b_i, s_i \geq 0 \tag{4}$$

$$\sum_{j=1}^{n} a_{ij} x_j = b_i \iff \sum_{j \in S} a_{ij} x_j + \sum_{j \notin S} a_{ij} (x_j^+ - x_j^-) = b_i \tag{5}$$

Outline

1 Linear Programming and Its Standard Form

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Reformulate NLP as LP

Many non-linear programming (NLP) problems can be transformed into linear programming.

- **Minimax/Maximin objective**:

$$\min_x \max_{i=1,\cdots,n} \left\{ c_i^\top x + d_i \right\} \Longrightarrow \min_{x,\Delta} \Delta, \Delta \geq c_i^\top x + d_i, \forall i$$

$$\max_x \min_{i=1,\cdots,n} \left\{ c_i^\top x + d_i \right\} \Longrightarrow \max_{x,\Delta} \Delta, \Delta \leq c_i^\top x + d_i, \forall i$$

- **Minimizing absolute values**:

$$\min_x \sum_{i=1}^{n} |x_i| \Longrightarrow \min_{x,t} \sum_{i=1}^{n} t_i, t_i \geq x_i, t_i \geq -x_i$$

Exercise 2

Given a vector $x \in \mathbb{R}^n$, a genearl definition of L_p-norm of x is given by:

$$||x||_p = (|x_1|^p + |x_2|^p + ... + |x_n|^p)^{1/p}$$

Formulate the following problems as LP(Linear Optimization Problem):

(1) $\min_x ||Ax - b||_1, \quad s.t. \quad ||x||_\infty \leq 1.$

(2) $\min_x ||x||_1 \quad s.t. \quad ||Ax - b||_\infty \leq 1.$

(3) $\min_x ||Ax - b||_1 + ||x||_\infty.$

Solution to Exercise 2 I

(1)

$$
\begin{aligned}
& \min_{x} ||Ax - b||_1, \quad s.t. \quad ||x||_\infty \leq 1, \\
\iff & \min_{x,y} 1^T y, \quad s.t. \quad ||x||_\infty \leq 1, \quad y \geq |Ax - b| \\
\iff & \min_{x,y} 1^T y, \quad s.t. \quad \max_{i \in [n]} \{x_i, -x_i\} \leq 1, \quad y \geq |Ax - b| \\
\iff & \min_{x,y} 1^T y, \\
& s.t. \quad x_i \leq 1, -x_i \leq 1, \forall i \in [n], \quad y \geq Ax - b, y \geq -Ax + b
\end{aligned}
$$

Solution to Exercise 2 II

(2)

$$
\begin{aligned}
& \min_{x} ||x||_1 \quad s.t. \quad ||Ax-b||_\infty \leq 1 \\
\iff & \min_{x,y} 1^T y, \quad s.t. \quad ||Ax-b||_\infty \leq 1, \quad y \geq |x| \\
\iff & \min_{x,y} 1^T y, \quad s.t. \quad \max_{i\in[n]}\{A_i x - b_i, b_i - A_i x\} \leq 1, \quad y \geq |x| \\
\iff & \min_{x,y} 1^T y, \\
& s.t. \quad A_i x - b_i \leq 1, b_i - A_i x \leq 1, \forall i \in [n], \quad y \geq x, y \geq -x
\end{aligned}
$$

Solution to Exercise 2 III

(3)

$$
\begin{aligned}
& \min_{x} ||Ax - b||_1 + ||x||_\infty \\
\iff & \min_{x,y,z} 1^T y + z, \quad s.t. \quad ||x||_\infty \le z, \quad y \ge |Ax - b| \\
\iff & \min_{x,y,z} 1^T y + z, \quad s.t. \max_{i \in [n]} \{x_i, -x_i\} \le z, \quad y \ge |Ax - b| \\
\iff & \min_{x,y} 1^T y + z, \\
& s.t. \quad x_i \le z, -x_i \le z, \forall i \in [n], \quad y \ge Ax - b, y \ge b - Ax
\end{aligned}
$$

Exercise 3

Formulate the next optimization problem as LP:

$$\min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y = |x|$$

where $x, y \in \mathbb{R}^n$, $A, B \in \mathbb{R}^{m \times n}$ and $|x| = (|x_1|, ..., |x_n|)^T$. Moreover, all the entries of B and d are positive.
(What if all entries of B and d are nonnegative instead of positive?)

Solution to Exercise 3 I

We have:

$$\min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y = |x| \tag{6}$$

$$\overset{why?}{\iff} \min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y \geq |x| \tag{7}$$

$$\iff \min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y \geq x, \quad y \geq -x$$

Solution to Exercise 3 II

$$\min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y = |x|$$

$$\overset{why?}{\Longleftrightarrow} \min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y \geq |x|$$

Optimal solutions and optimal values for (6) and (7) are the same.
Suppose on the contrary the optimal solution (x^*, y^*) for (7) is not optimal to (6), i.e. $\exists y_i^* > |x_i^*|$ for some i. Note that $(x^*, |x^*|)$ is a feasible solution by the positivity of B :

$$Ax + B|x^*| < Ax + By^* \leq b.$$

Solution to Exercise 3 III

$$\min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y = |x|$$

$$\overset{why?}{\Longleftrightarrow} \min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y \geq |x|$$

Moreover, its objective value will be smaller than that of (x^*, y^*) by the nonnegativity of d

$$c^\top x^* + d^\top |x^*| < c^\top x^* + d^\top y^*,$$

which contradicts with the optimality of (x^*, y^*). Thus, the optimal value of y should be $|x^*|$.

Solution to Exercise 3 IV

$$\min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y = |x|$$

$$\overset{why?}{\iff} \min_{x,y} c^T x + d^T y, \quad s.t. \quad Ax + By \leq b, \quad y \geq |x|$$

What if we only have the nonnegativity rather than positivity for d and B?

- The optimal values for (6) and (7) are still the same by following the similar argument.
- The optimal solutions for (6) and (7) are not necessarily the same. For instance, consider the case where d and B are zero, then we can have arbitrarily large y without changing the optimal value.

Summary

After this tutorial, you should know

- What's the standard form of LP? How to transform any LP into its standard form?
- Common tricks for transforming NLP into LP, such as minimax objective problem and minimizing absolute value problem.

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